Chapter -4

- **Characteristics of computer memory systems**

Location

Memory can be internal (e.g., registers, main memory, cache) or external (e.g., hard drives, optical disks).

Capacity

Capacity is the amount of data memory can hold, measured in words or bytes.

Unit of Transfer

The unit of transfer is the amount of data moved in or out at once; can be a word, block, or byte.

Access Method

Four main access methods: Sequential (must go through data in order), Direct (can skip to approximate location), Random (any location accessed directly), and Associative (match by content, not address).

Performance

Performance is judged by access time (how fast data is read/written), memory cycle time, and transfer rate (speed of moving data).

Physical Type and Characteristics

Memory can be semiconductor, magnetic, optical, or magneto-optical; it may be volatile/nonvolatile or erasable/nonerasable.

Organization

Memory is often organized into modules for easier management and expansion.

- **Elements of cache design**

Cache addresses: Can be logical or physical; determine how memory locations are mapped.

Cache size: Total storage capacity of the cache.

Mapping function: Methods for placing memory blocks (direct, associative, set-associative) in cache.

Replacement algorithm: Strategy for choosing which cache line to overwrite (LRU, FIFO, LFU, random).

Write policy: How changes are written—write through, write back, or write once.

Line size: The size of each block/data line in the cache.

Number of caches: Can be single level, two levels, unified, or split depending on system architecture.

- **Cashe/Main Memory Structure**

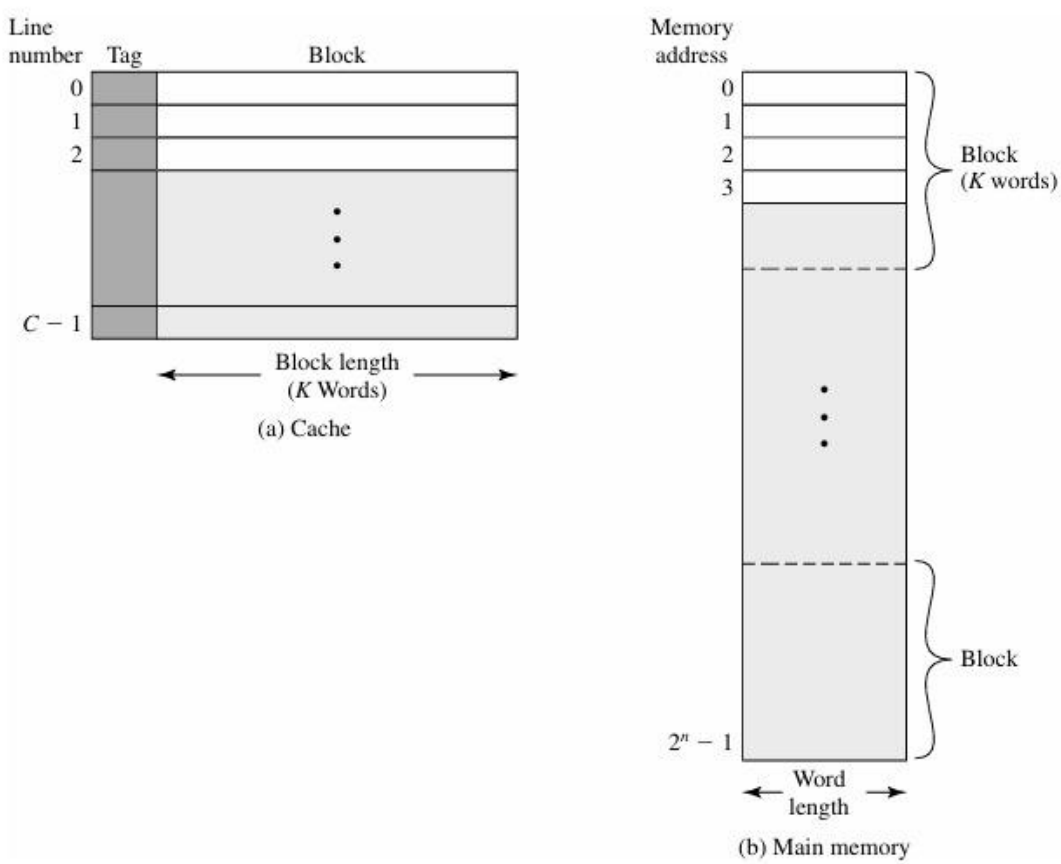
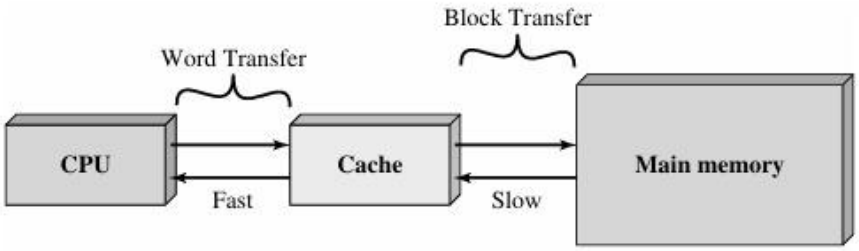
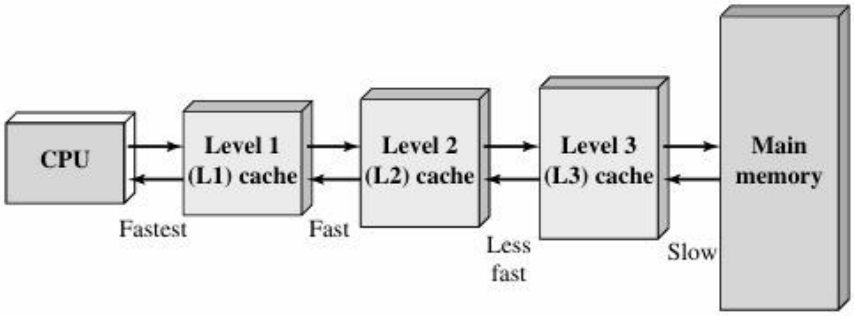


Figure 4.4 Cache/Main Memory Structure

- **Cashe Organazation**



(a) Single cache



(b) Three-level cache organization

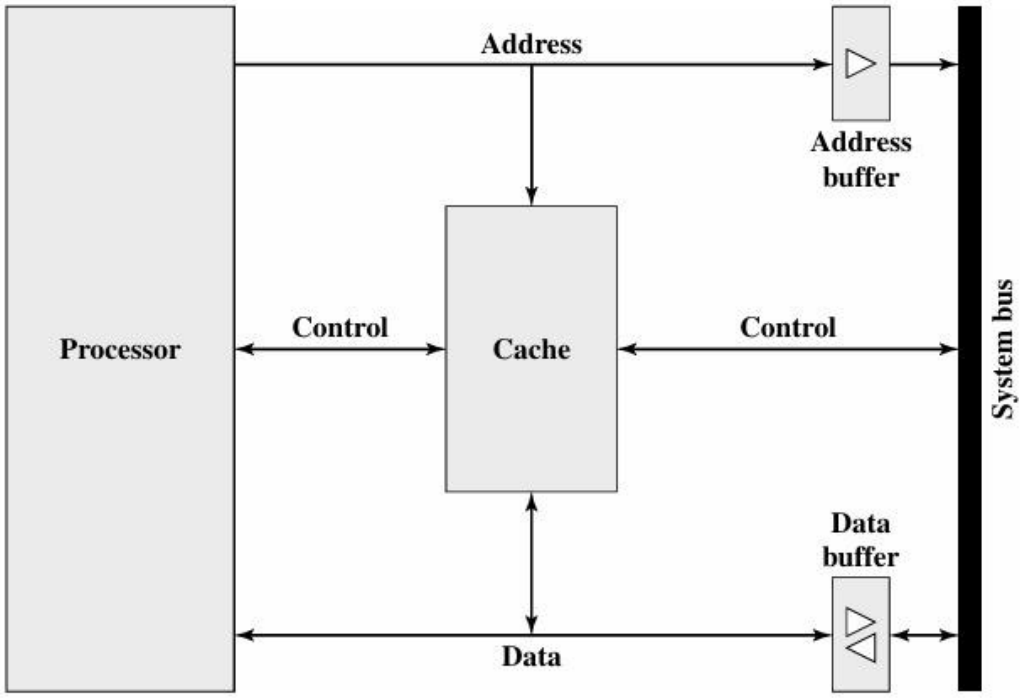


Figure 4.6 Typical Cache Organization

4.1 What are the differences among sequential access, direct access, and random access?

Sequential access: Data is accessed strictly in order; to read a specific record, all previous records must be checked (e.g., magnetic tape).

Direct access: Data is accessed by jumping to an approximate location, then sequentially searching for exact information (e.g., disk drives).

Random access: Any memory location can be accessed directly and instantly, regardless of previous accesses (e.g., RAM).

4.2 What is the general relationship among access time, memory cost, and capacity?

Faster access time generally means higher cost per bit and lower capacity.

Slower memory (like disk) is cheaper and can offer large capacity, but access time is much longer.

Faster memory (like cache or RAM) costs more and is provided in smaller amounts.

4.3 How does the principle of locality relate to the use of multiple memory levels?

The principle of locality states that programs tend to access a relatively small portion of their address space at any moment.

Multiple memory levels (e.g., cache, main memory, disk) exploit locality by keeping frequently used data in faster, smaller memory and less-used data in slower, larger storage.

4.4 What are the differences among direct mapping, associative mapping, and set-associative mapping?

Direct mapping: Each block maps to exactly one cache line.

Associative mapping: A block can be placed in any line in the cache.

Set-associative mapping: A block maps to a set of lines (e.g., 2 or 4); can go anywhere within its set.

4.5 For a direct-mapped cache, a main memory address is viewed as consisting of three fields. List and define the three fields.

Tag: Identifies the stored data to distinguish between memory blocks that map to the same cache line.

Line number (index): Selects which cache line a memory block maps to.

Byte/word offset: Specifies the exact byte/word within the cache line.

4.6 For an associative cache, a main memory address is viewed as consisting of two fields. List and define the two fields.

Tag: Uniquely identifies data stored in the cache.

Byte/word offset: Specifies the exact location within the block.

4.7 For a set-associative cache, a main memory address is viewed as consisting of three fields. List and define the three fields.

Tag: Identifies the block uniquely for searching within the set.

Set index: Selects which set the memory block should map to.

Byte/word offset: Exact position within the block/set line

4.8 What is the distinction between spatial locality and temporal locality?

Temporal locality: Refers to reuse of specific data or resources within a short period of time (same data accessed repeatedly).

Spatial locality: Refers to use of data elements within relatively close storage locations (addresses near each other accessed together).

4.9 In general, what are the strategies for exploiting spatial locality and temporal locality?

Spatial locality strategies: Use larger blocks or prefetching to bring adjacent data into fast memory when one item is accessed.

Temporal locality strategies: Use caching to keep recently accessed data in fast memory so it can be quickly reused.